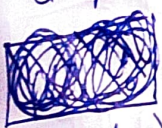


$\therefore$ We shall prove $\neg((s \wedge t) \vee (u \wedge v)) \rightarrow \neg(p \wedge q \wedge r)$ is True.

Let, $\neg((s \wedge t) \vee (u \wedge v))$ be True. By De Morgan's Law, $\neg(s \wedge t) \wedge \neg(u \wedge v)$ is True. By De Morgan's Law, $(\neg s \vee \neg t) \wedge (\neg u \vee \neg v)$ is True. By Distributive Law, $((\neg s \vee \neg t) \wedge \neg u) \vee ((\neg s \vee \neg t) \wedge \neg v)$ is True. By Distributive Law, $(\neg s \wedge \neg u) \vee (\neg t \wedge \neg u) \vee (\neg s \wedge \neg v) \vee (\neg t \wedge \neg v)$ is True. By Defn. of OR, at least one of $(\neg s \wedge \neg u)$, $(\neg t \wedge \neg u)$, $(\neg s \wedge \neg v)$, $(\neg t \wedge \neg v)$ is True.

Case 1: Let, $\neg s \wedge \neg u$ be True. By Defn. of AND, $\neg s$ is True and $\neg u$ is True. $\therefore m \neq 1$ and $m \neq -1$

If $m = 0$, $mn = 0$. $\therefore mn \neq 1$. $\therefore \neg p$ is True. By Defn. of negation, $\neg(\neg p)$ is False. By Double negation Law, p is False. By defn. of AND, $(p \wedge q \wedge r)$ is False. By defn. of negation, $\neg(p \wedge q \wedge r)$ is True.

 If m is not an integer, then $\neg q$ is True. By Defn. of negation, $\neg(\neg q)$ is False. By Double negation Law, q is False. By defn. of AND, $(p \wedge q \wedge r)$ is False. By defn. of negation, $\neg(p \wedge q \wedge r)$ is True.

If m is an integer, then if $mn = 1$, then $n = \frac{1}{m}$ which is not an integer. $\therefore \neg p$ is True. By defn. of negation, $\neg(\neg p)$ is False. By double-negation Law, p is False. By defn. of AND, $(p \wedge q \wedge r)$ is False. By defn. of negation, $\neg(p \wedge q \wedge r)$ is True.

If $mn \neq 1$, then $\neg p$ is True. By defn. of negation, $\neg(\neg p)$ is False. By double-negation Law, p is False. By defn. of AND, $(p \wedge q \wedge r)$ is False. By defn. of negation, $\neg(p \wedge q \wedge r)$ is True.

Case 2: Let, $\neg t \wedge \neg v$ be True. By defn. of AND, $\neg t$ is True and $\neg v$ is True. $\therefore n \neq 1$ and $m \neq -1$.